



JURONG JUNIOR COLLEGE

JC2 Preliminary Examination 2018

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

10 September 2018

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
Section B	
Total	

This document consists of **19** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

- Over thousands and millions of years, there have been natural cycles in the Earth's climate. There have been ice ages (e.g. the Quaternary Ice Age) and warmer interglacial periods. Climatic changes can be investigated using evidence left in tree rings, layers of ice in glaciers, ocean sediments and layers of sedimentary rocks. For example, bubbles of air in glacial ice trap tiny samples of Earth's atmosphere, giving scientists a history of greenhouse gases that stretches back more than 800,000 years.

These changes in climate affect plant distribution and physiology. Fig. 1.1 is a diagram showing the topographical profile of two mountains in the tropics during a warm phase and a cool phase in the Earth's climate. The shape of the lines corresponds to a vertical section through the mountains to show their height and shape. The distribution of rain forest vegetation is also shown.

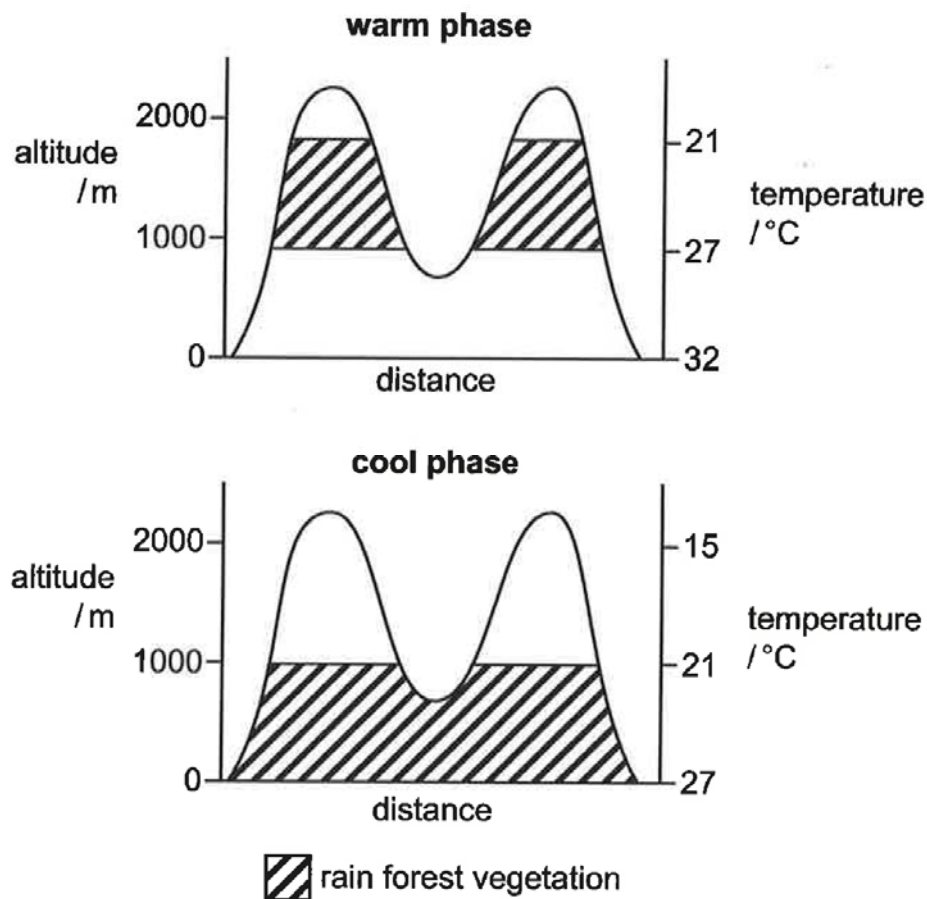


Fig. 1.1

- (a) (i)** Describe and explain the effect of climate change on the distribution of rain forest vegetation in the tropics, as shown in Fig. 1.1. [4]

- (ii)** Over millions of years there are repetitive cycles of climate change resulting in a pattern of alternating warm and cool phases in the Earth's climate.

Suggest how repeated changes in climate between the two phases shown in Fig. 1.1 may lead to evolution of new species and greater species diversity. [4]

- (iii) As the atmosphere is thin at the top of the mountains, the plants growing at this region absorb high amounts of light energy from the sun since the sun rays are not scattered or reflected by the air molecules.

Explain how photosynthesis may be affected. [3]

- (b) Changes in the Earth's climate are also expected to have an impact on insects such as the *Aedes aegypti* mosquitoes which are commonly found in the tropics and sub-tropics. As the mosquitoes are the vectors carrying dengue virus and causes dengue in humans, their life-cycles have been studied in detail to manage their population size.

Describe the life-cycle of *Aedes aegypti*. [4]

- (c) Another mosquito-borne disease of concern worldwide is malaria. Malaria was common in Italy, a European country situated in the Northern Hemisphere. Widespread land drainage together with the use of the insecticide DDT and the drug chloroquine eradicated both the mosquito vectors and the malaria parasites, *Plasmodium*, in the 1950s. Due to the success of these measures, they were later discontinued.

Articles in the scientific literature more recently show that malarial mosquitoes are returning to Italy and increasing their numbers and their northerly range, with some cases of malaria being reported. In general, winters are milder and summers hotter in the south of the country, with temperatures decreasing in the north, especially in winter.

Discuss whether the return of malaria to Italy can be attributed to climate change. [4]

- (d) In many parts of the world, the malaria parasites, *Plasmodium*, have become resistant to chloroquine. In these areas, one of the most effective anti-malarial drugs currently in use is artemisinin. Artemisinin works by binding to an enzyme in *Plasmodium* called PfATP6, acting as an inhibitor.

A substance called curcumin, which has long been used as a spice and yellow food colouring in India and other countries, is also known to act against chloroquine-resistant *Plasmodium*. A group of researchers predicted that curcumin acts by binding to the same enzyme as artemisinin.

In order to test this hypothesis, and to try to find other similar substances that might work better than curcumin, the researchers used theoretical modelling to:

- look at the chemical structures of various molecules with a similar structure to curcumin (curcumin analogues)
- generate a three-dimensional model of the structure of the enzyme PfATP6
- investigate whether each curcumin analogue could bind to PfATP6.

The researchers predicted that several of the curcumin analogues would bind more strongly than curcumin to PfATP6.

- (i) Suggest advantages of using theoretical models in this research, rather than testing possible drugs in the laboratory. [2]

- (ii) Suggest why theoretical modelling cannot completely replace laboratory trials in the search for new drugs. [2]

(e) In places such as Africa where malaria is endemic, sickle cell anaemia — an inherited blood disorder — is prevalent in the populations. Sickle cell anaemia is caused by a single base pair substitution.

(i) Describe the effects of the mutation of the gene involved in sickle cell anaemia. [4]

(ii) Only those individuals that inherit two copies of the recessive allele which causes sickle cell anaemia develop the disease. If left untreated, these individuals have a shorter than normal life expectancy. As such, it would be expected that this mutation would be rare in human populations. However, observations made during the mid-20th century revealed that there are sometimes 10-40% of the population carrying this recessive allele.

Explain how there can be a high proportion of the population carrying the recessive allele even when it is disadvantageous. [3]

[Total: 30]

- 2 Fig. 2.1 shows stem cells in bone marrow dividing by mitosis to form blood cells. Each time a stem cell divides, it forms a replacement stem cell and a cell that develops into a blood cell.

Stem cells in the bone marrow give rise to reticulocytes, phagocytes, B-cells and T-cells.

Reticulocytes will differentiate into red blood cells.

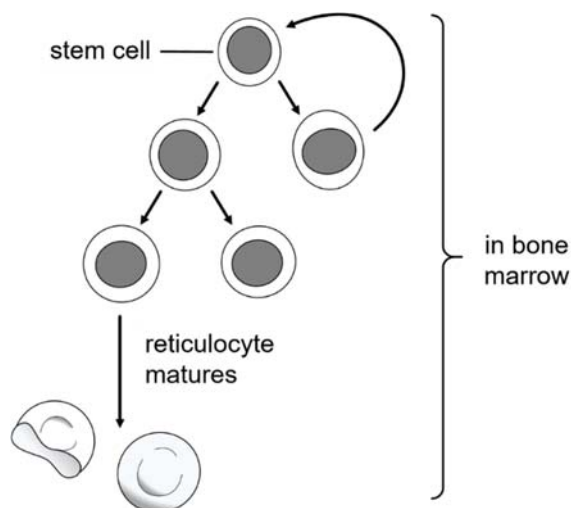


Fig. 2.1

- (a)** Describe how a functional red blood cell develops from a stem cell. [3]

- (b)** Stem cells share some features of cancer cells, such as having long life span and ability to replicate for extended periods of time.

In some cases, the use of stem cells in medical treatment of genetic diseases appears to increase the risk of cancer.

Suggest why there might be a connection between the medical use of stem cells in treatment and increased cancer risk. [2]

Stem cells can also differentiate to form T cells. T cells are arguably the most important cells in adaptive immunity, as they are required for almost all adaptive immune responses.

A person infected with the HIV virus was monitored for several years for the level of T cells and HIV particles.

The results are summarised in Fig. 2.2.

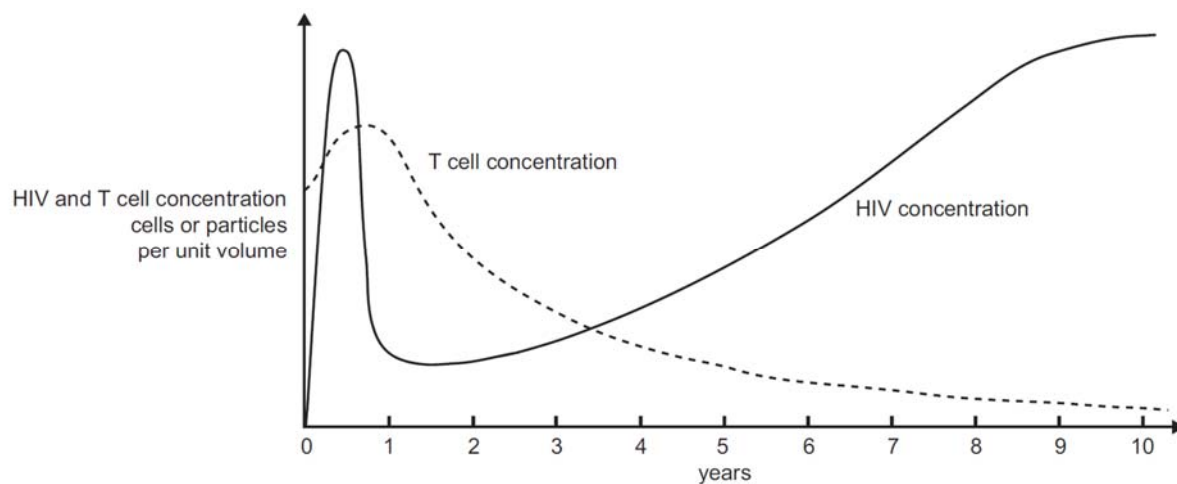


Fig. 2.2

(c) Explain the trend observed from 0.5 to 10 years. [4]

Enzymes essential in the life cycle of human immunodeficiency virus (HIV) have been at the focus of research. HIV-1 protease is one such enzyme.

HIV-1 protease is formed from two identical polypeptides, each of which is made up of 99 amino acids. The polypeptides both have α helices and β strands. The active site of the enzyme is formed by three amino acids from each polypeptide. Fig. 2.3 shows HIV-1 protease.

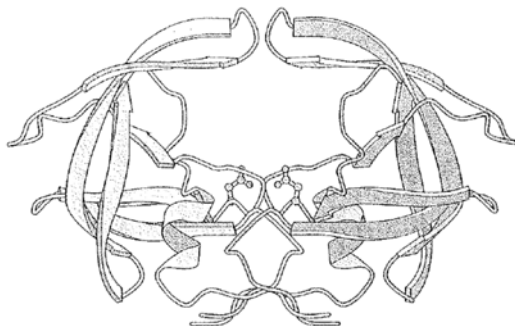


Fig. 2.3

Over the years, knowledge of the structure led to designs of HIV protease inhibiting drugs with improved antiviral properties. Despite the qualified success of these inhibitors, the high mutation rate associated with RNA viruses continues to hamper the long-term clinical efficacy of HIV protease inhibitors.

- (d)** Suggest why this means that researchers may need to continue to develop new drugs in order to prevent the action of HIV-1 protease. [2]

[Total: 11]

- 3** G protein-coupled receptors (GPCRs) constitute a large protein family of receptors that detect molecules outside the cell and activate internal signal transduction pathways and, ultimately, cellular responses in mammals.

(a) Describe two ways in which GPCRs are similar in structure and explain why these features are important for their function. [4]

The glucagon receptor is a Class B GPCR that plays a central role in the regulation of blood glucose levels and glucose homeostasis. The actions of glucagon are transduced via activation of the glucagon receptor.

(b) Describe how binding of glucagon aids in the maintenance of a constant blood glucose level. [4]

(c) The following findings were concluded from studies on the expression of the glucagon receptor mRNA in liver, adipose tissues and pancreatic islets.

- The promoter of the glucagon receptor gene contain regulatory elements for glucose and cyclic AMP.
- The expression of the glucagon receptor mRNA is stimulated by glucose and inhibited by cyclic AMP in liver.
- The number of glucagon receptors expressed in the liver can be regulated by glucagon.

Comment how the above features are advantageous to mammals. [1]

[Total: 9]

Section B starts on page 14

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

- 4 **(a)** The organelles of the endomembrane system in eukaryotic cells are related through direct contact or by the transfer of membrane segments as vesicles.

Outline the functions of the organelles of the endomembrane system and state the structural similarities between these organelles. [15]

- (b)** Binary fission is a process where a single bacterial cell divides into two genetically identical daughter cells. Discuss if human cells are genetically identical and contrast the organisation of bacterial and human genome. [10]

[Total: 25]

- 5 **(a)** Variation exists in individuals of the same species in a population due to a number of different reasons. Describe what causes variation and why it is important in natural selection. [15]

- (b)** Discuss, using known examples, how limiting factors can influence the rate of various biological processes. [10]

[Total: 25]

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